

Krishna's Set using DL, NLP, ML course

①

Set - 27

Ridge Lasso > Elastic net
ML algorithms

Let's :-

Ridge Regression

Let's :-

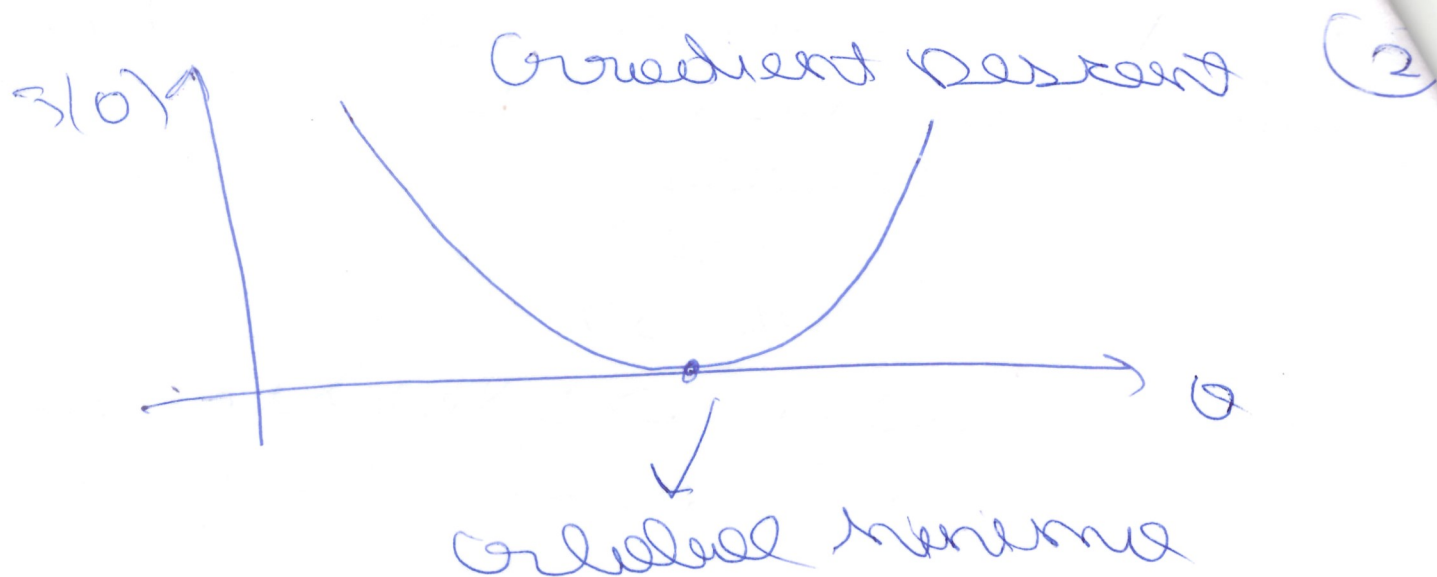
Lasso > Elastic Net ML

Ridge Regression, Lasso Regression,
Elastic Net Regression

Linear regression



Independent features



Cost function:

$$= \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

① Ridge regression

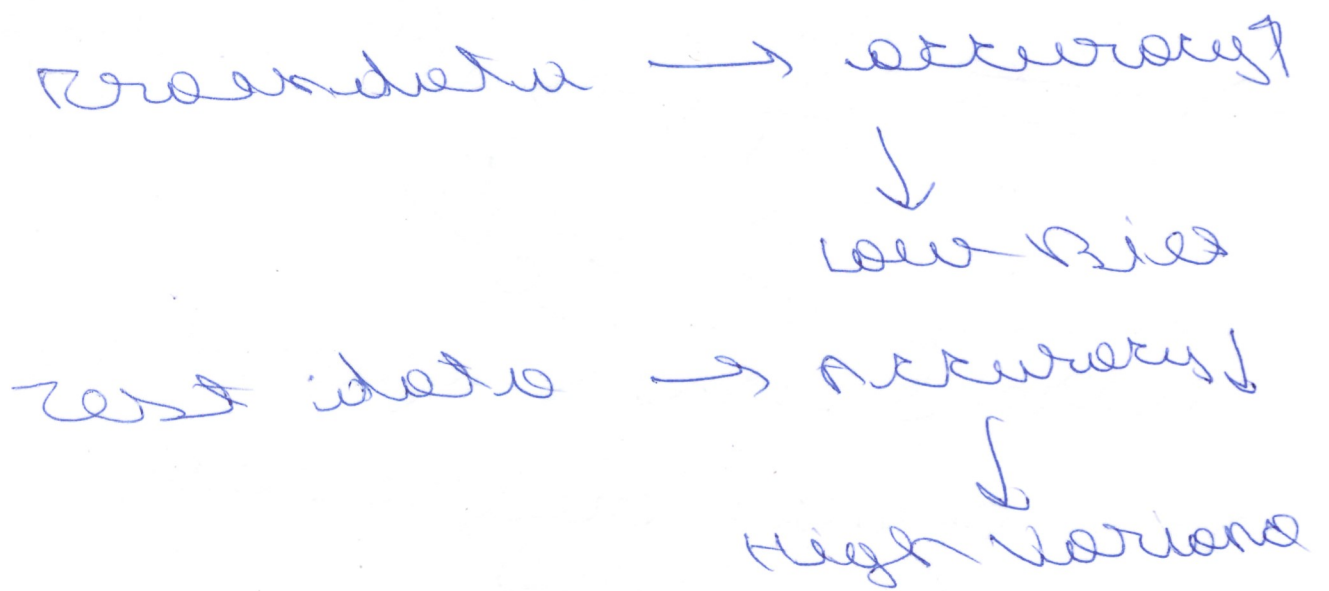
Suppose we have a single
independent feature

we have 2 specific
training points in data
set

⇒ Suppose, we have 2
data points on the
graph

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1) More samples, if we
add more training, then
we get the condition
of overfitting.
- overfitting -



• Ridge regression is also
known as L_2 regulari-
zation

- This regularization
is used to reduce
overfitting.

• Basically Ridge Regression models or higher parameter turning off linear regression.

(*) Cost function:

↳ for Ridge regression

$$\Rightarrow \frac{1}{2m} \sum_{i=1}^m (\text{model}^{(i)} - y^{(i)})^2$$

⇓

- 0

Here this is equal to 0

for Ridge regression we use that -

$$\Rightarrow \frac{1}{2m} \sum_{i=1}^m (\text{model}^{(i)} - y^{(i)})^2$$

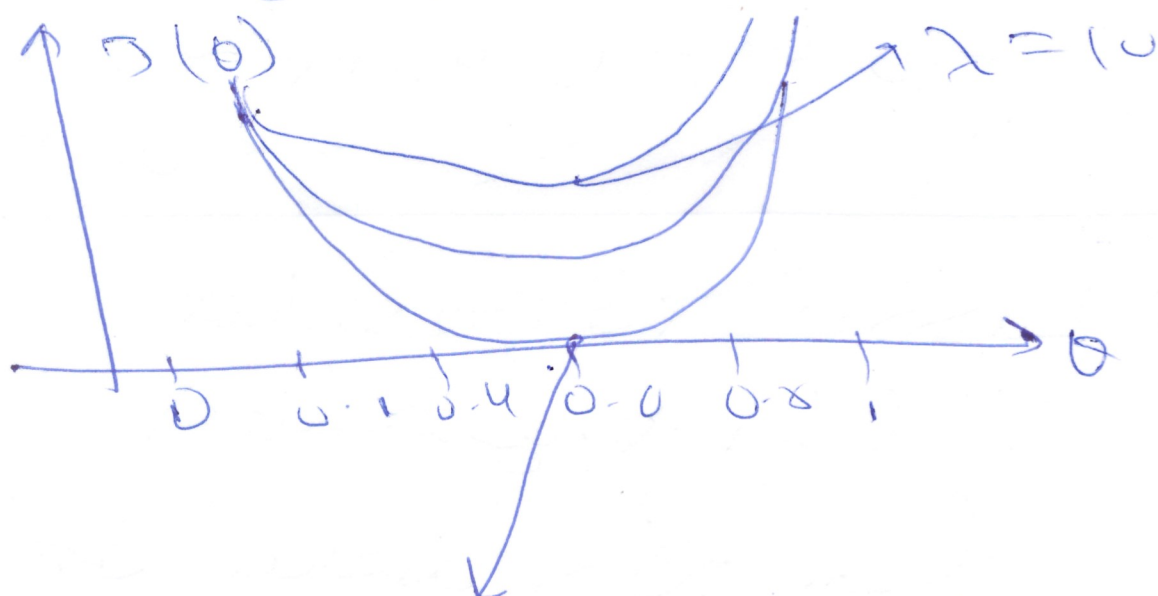
⇓

$$+ \lambda \sum_{i=1}^m (\text{slope})^2$$

5) Suppose if a question is asked that:

→ Relation b/w λ & (slope)²

→ Let's draw the graph as:-



Global minimum

$$\lambda = 0$$

$$J_2(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3$$

(6)

$$h_0(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3$$

$$= 0.34 + 0.52x_1 + 0.48x_2 + 0.24x_3$$

$$= 0.34 + 0.40x_1 + 0.38x_2 + 0.14x_3$$

(2) Lasso Regression (L_1 regression)

→ It's very important that we include variables ourselves

cost function =

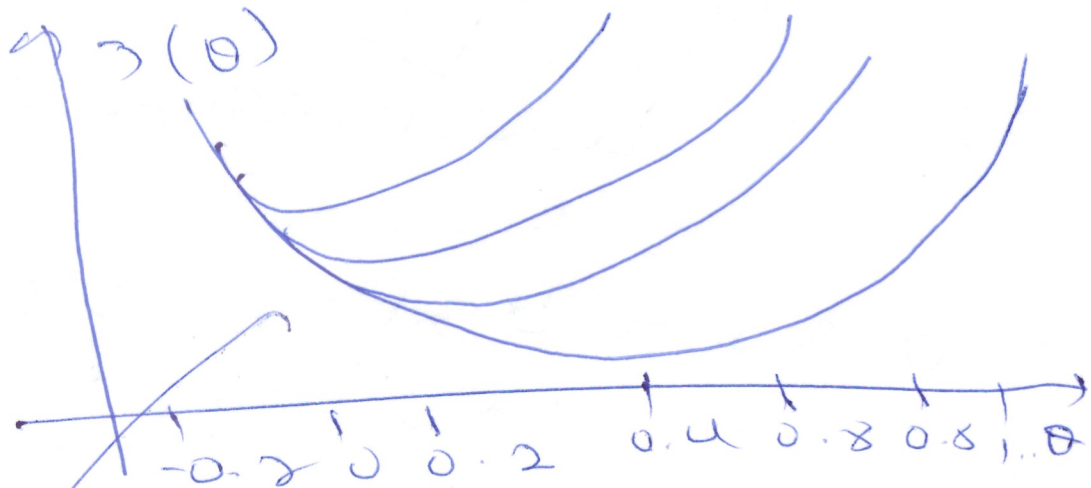
$$\frac{1}{2m} \sum_{i=1}^m (h_0(x^{(i)}) - y^{(i)})^2 + \lambda \sum_{i=1}^m |slope|$$

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Here we will

$$x \sum_{i=1}^n |s_i|$$

It can be represented
as graph as



This is gradient descent
curve with $\gamma = 0$

After one point of
time, θ value

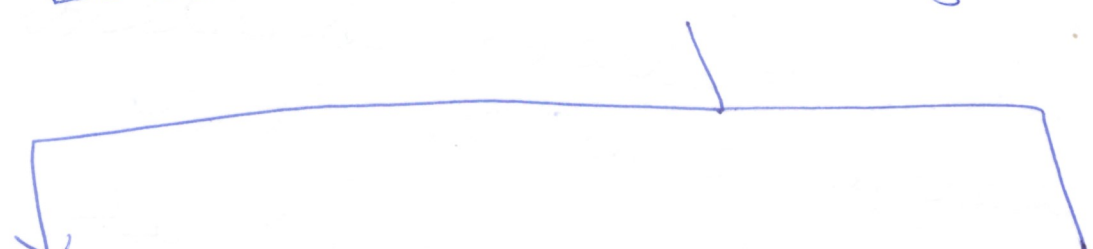
$$h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \theta_4 x_4$$

$$= 0.52 + 0.65x_1 + 0.72x_2 + 0.34x_3 + 0.12x_4$$


 Lasso regression

$$= 0.52 + 0.51x_1 + 0.60x_2 + 0.14x_3 + 0x_4$$

③ Elastic Net Regression


 ① Reduce overfitting

Cost function follows as:-

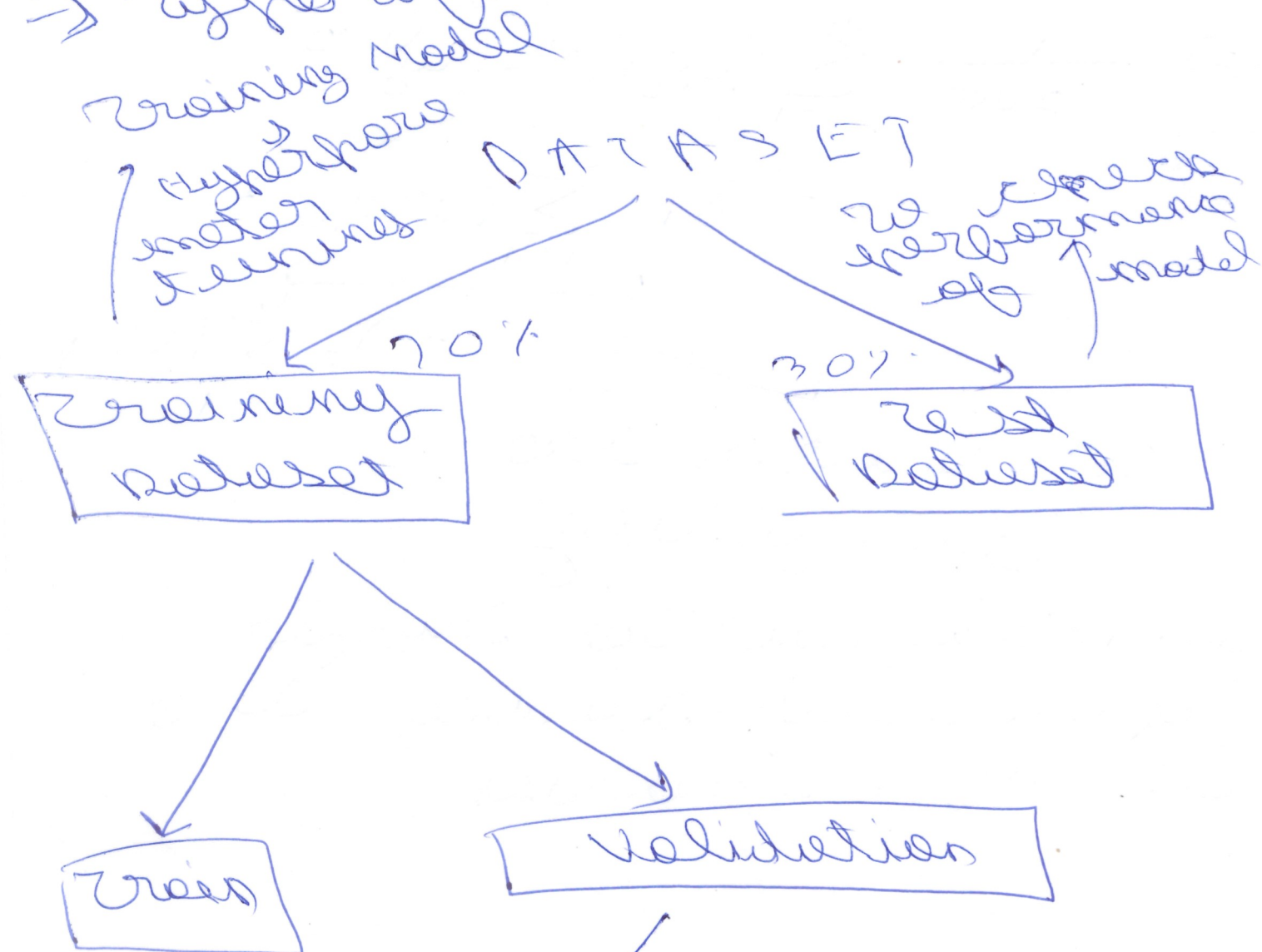
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$$= \frac{1}{2m} \sum_{i=1}^m (y_{obs}^i - y_{pred}^i)^2 + \lambda_1 \sum_{i=1}^m (\text{slope})^2 + \lambda_2 \sum_{i=1}^m |\text{slope}|$$

It reduces the
overfitting

Basically, we're regularizing
the linear regression
parameter tuning the

⇒ types of cross validation



We perform the Hyperparameter tuning with validation data set.

There are different types of cross validation methods we use:-

① Leave one out CV (LOOCV)

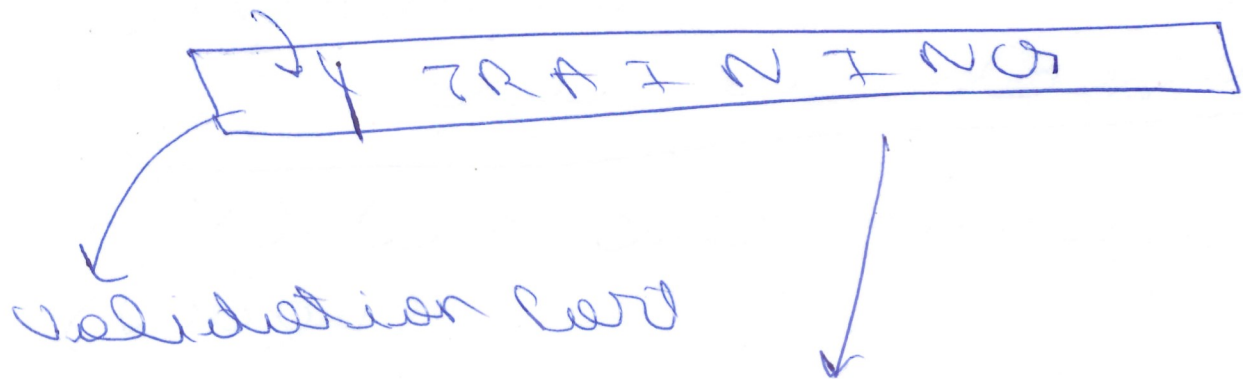
• In the training data the no. of records are 500.

Training → 500 Records

↓
we split this data into training & validation data.

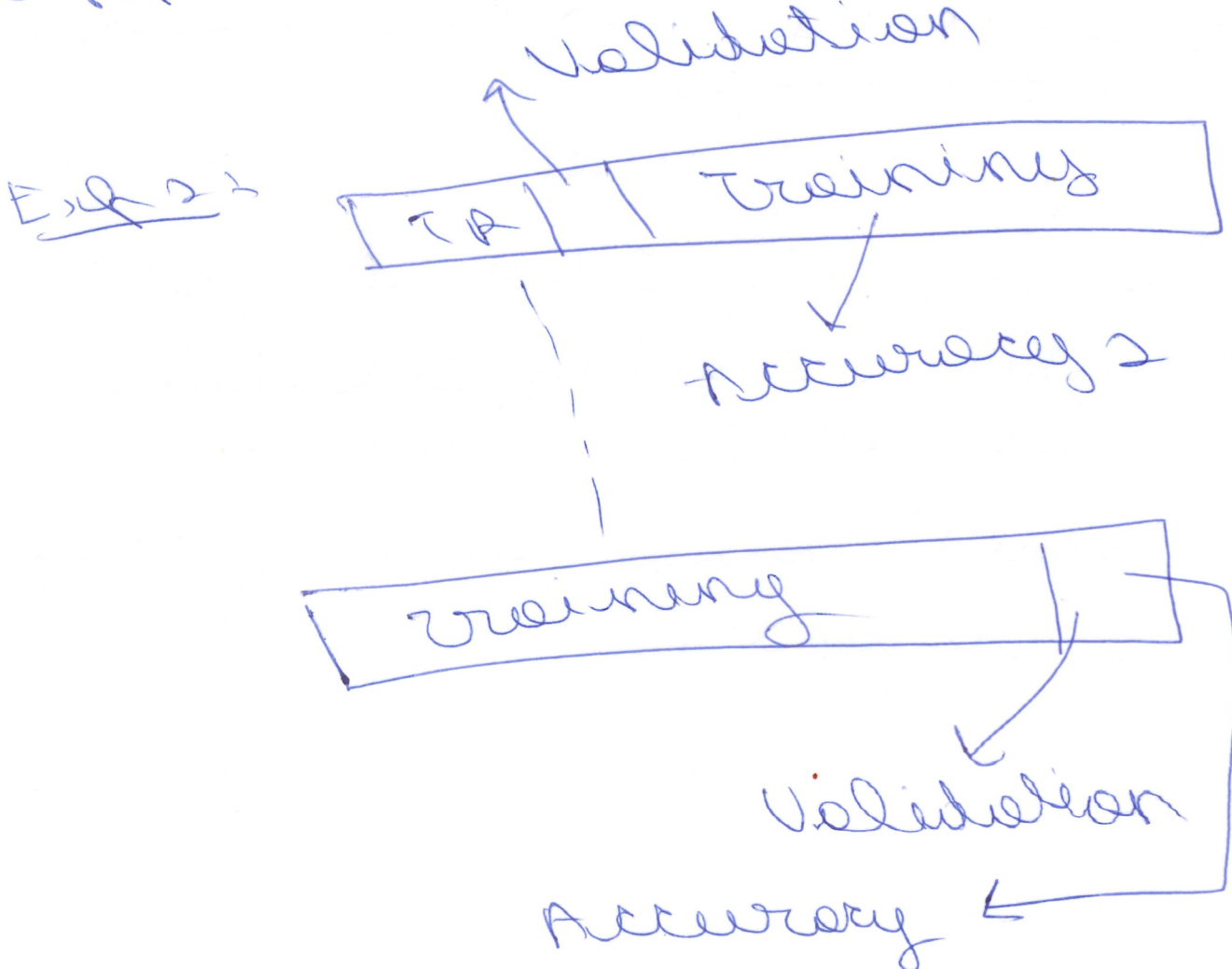
• we will be taking the 500 records.

for the experiment 1.



We will get some accuracy based on the validation set.

Suppose in Experiment 2



Example

As the dataset size increases, complexity of learning the model, also increases.

Overfitting $\rightarrow$ Training $\uparrow$

accuracy is very high $\uparrow$

Since the validation set is very small, accuracy is

decreasing.

when we test the model with new test data



the model performance will be lower

~~2nd~~ 2nd second type of cross validation is:-

(2) Leave P out CV:-

Here we set P value & select n as higher parameter tuning.

(3) 2nd 3rd type is K-Fold CV

$$K = 5$$

$$n = 500$$

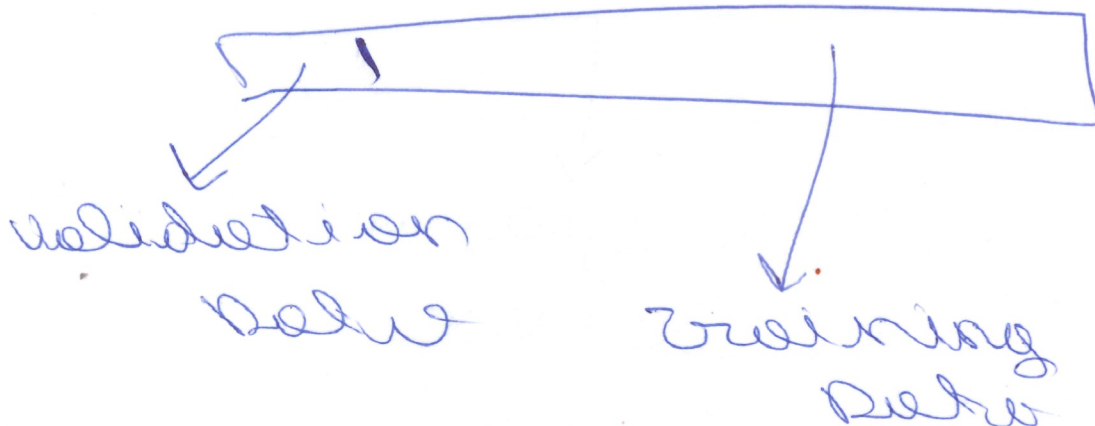


↓

$$CV = \frac{500}{5} = 100$$

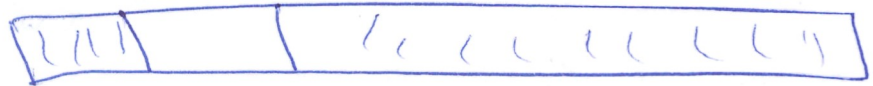
Test size

(CV = 1 ~~2nd~~ Ex 1)

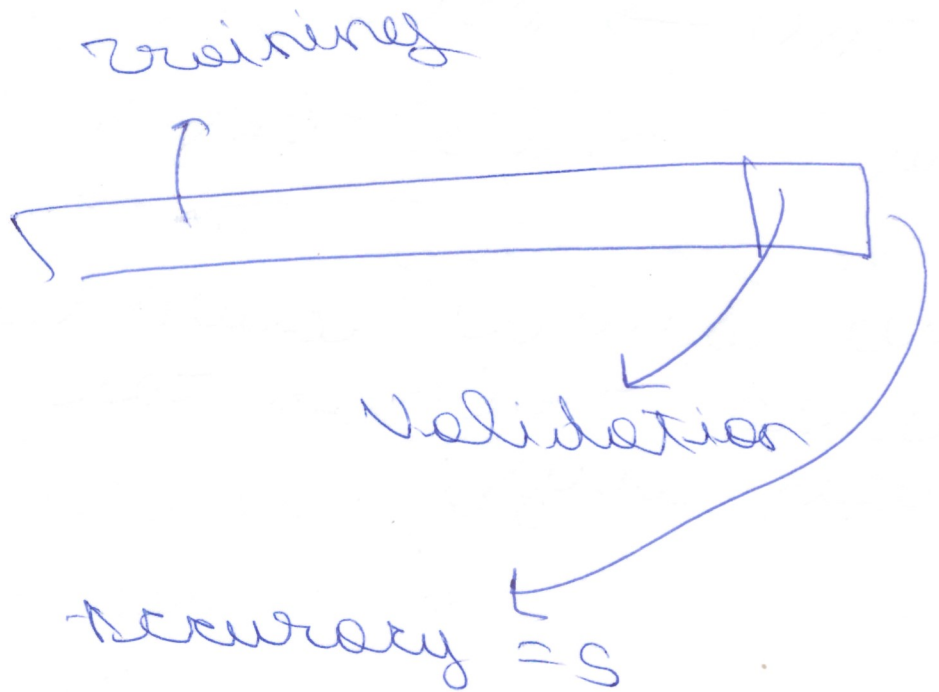


the next 500 records
we see it as:-

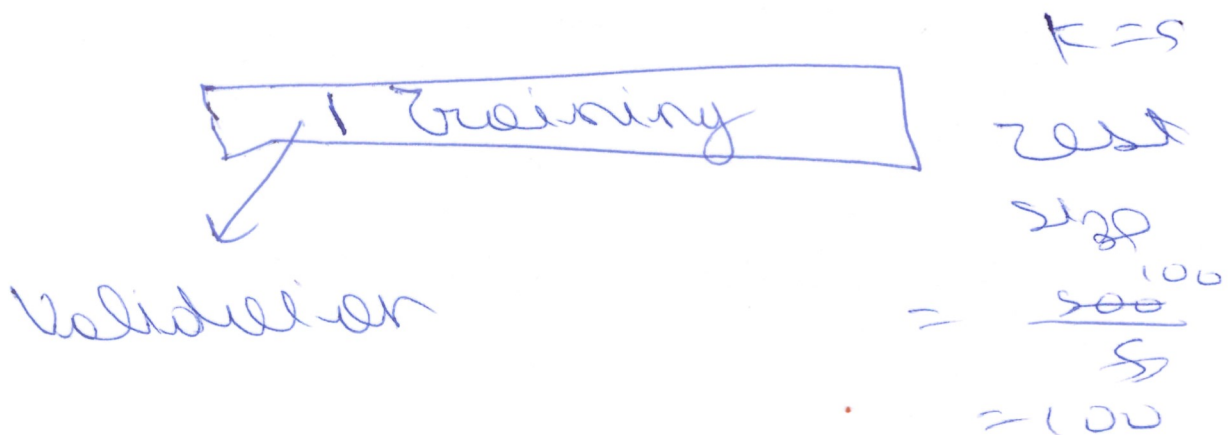
Ex 2:



Ex 5:



④ Stratified K Fold CV:-



⑤ time series CV

⑩

Product sentiment Analysis

time

JAN $\rightarrow$ DEC

Day 1 Day 2 Day 3 Day 4 ... Day N

- we divide our dataset
into train $\rightarrow$ test based
on days.